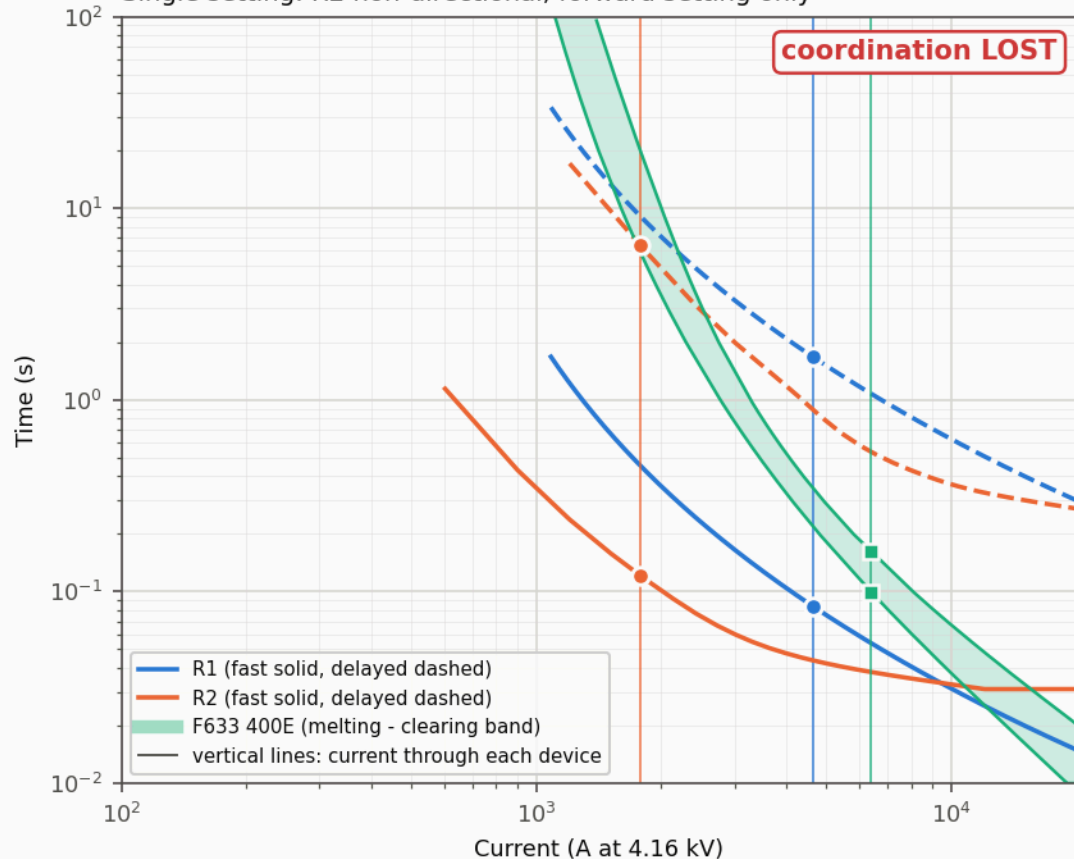
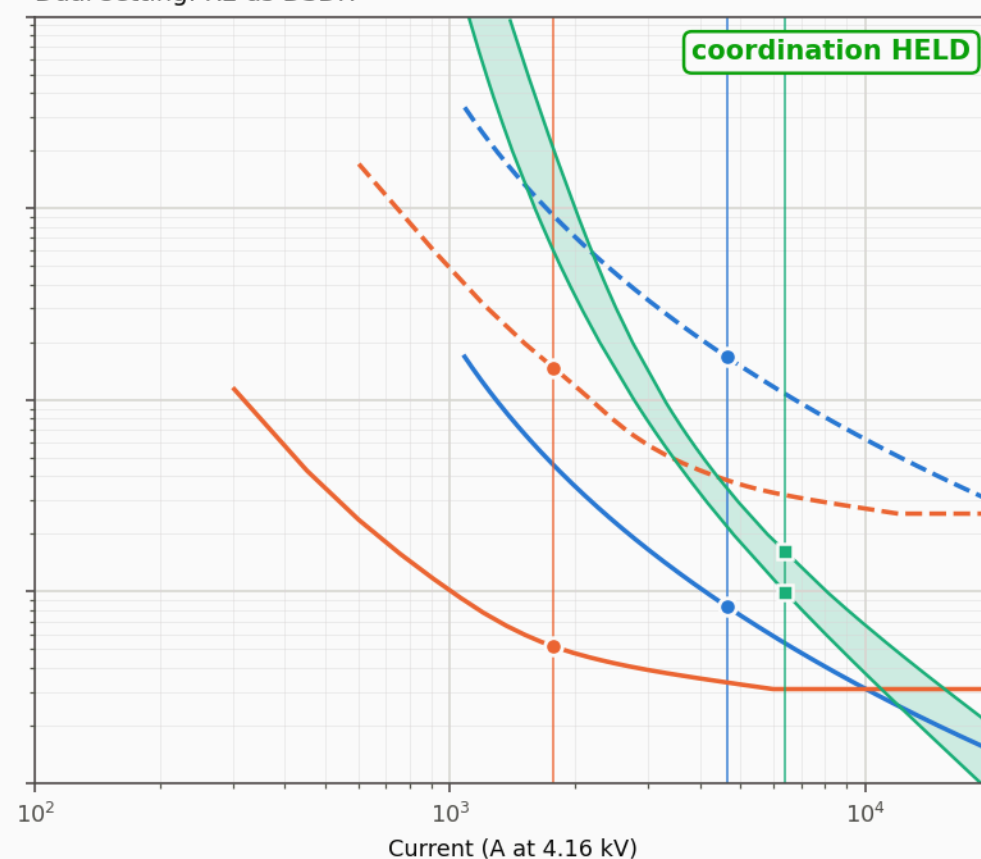


Case 05 - 3-phase fault at 10 % of 632-633, $Z_f = 0$ ohm (bolted), DG 4.05 MVA connected at 692 (Fig. 13)

Single setting: R2 non-directional, forward setting only



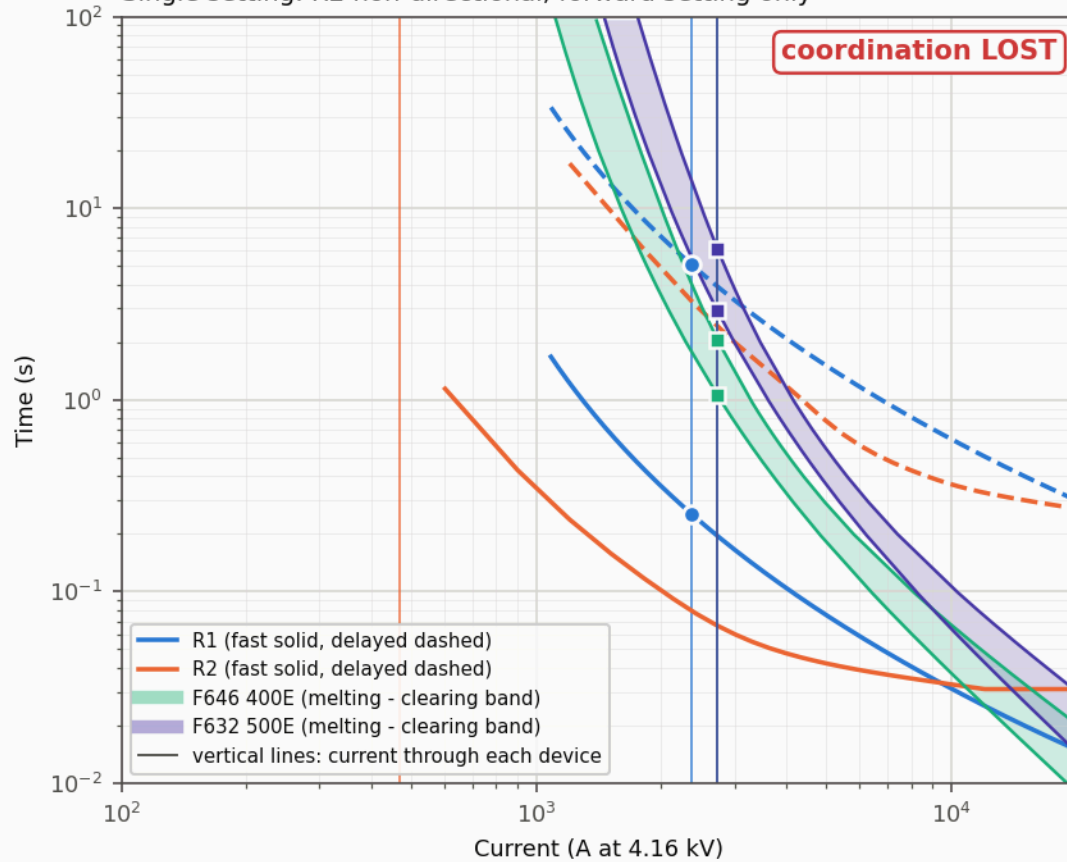
Dual setting: R2 as DSDR



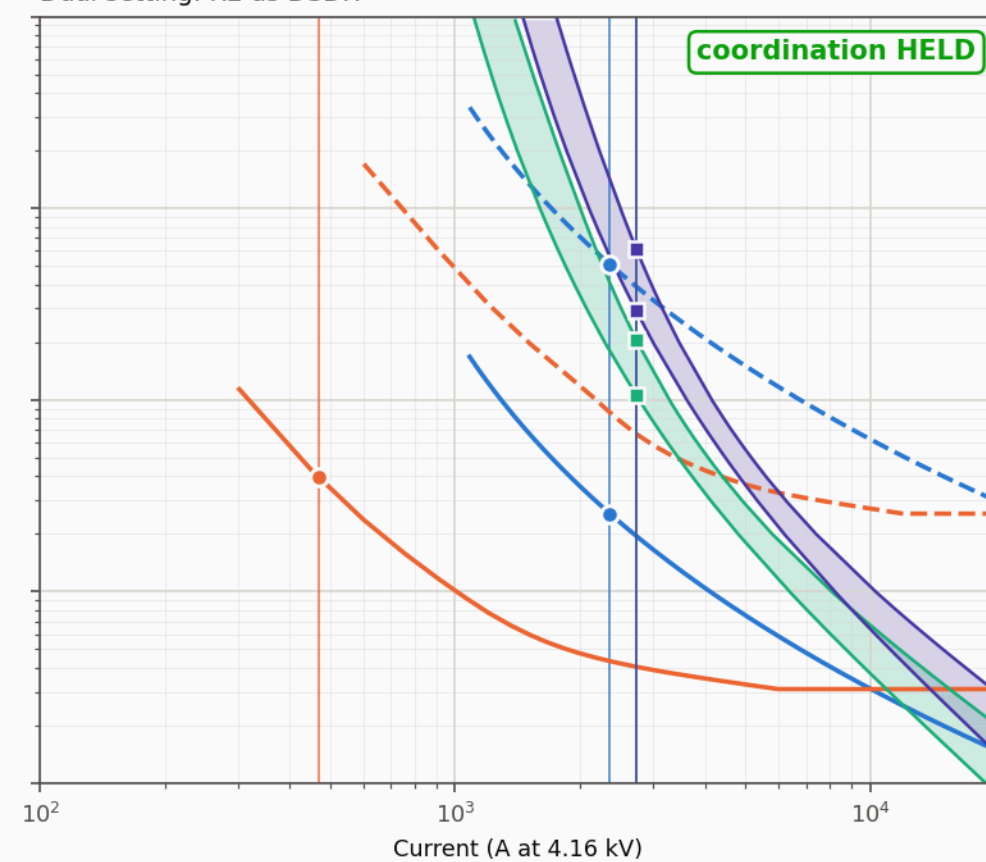
Quantity	Single setting	Dual setting (DSDR)
3-phase fault at 10 % of 632-633	LLL, phases a-b-c	(same fault and currents)
Fault impedance Z_f	0 ohm (bolted)	(same fault and currents)
Fault-path current (F633)	6398 A	(same fault and currents)
Current through R1, direction	4645 A, forward (RG60 -> 632), grid contribution	(same fault and currents)
Current through R2, direction	1777 A, REVERSE (671 -> 632), DG contribution only	(same fault and currents)
R2 setting applied (fast / delayed)	FORWARD: plug 300/600 A, starts 600/1200 A, TMS 0.1/1.0	REVERSE: plug 150/300 A, starts 300/600 A, TMS 0.1/1.0
R1 fast / delayed	0.084 s / 1.672 s	0.084 s / 1.672 s
R2 fast / delayed	0.121 s / 6.426 s	0.052 s / 1.470 s
Primary fuse melts / clears	F633 400E: 0.100 s / 0.163 s	F633 400E: 0.100 s / 0.163 s
Backup fuse melts	-	-
Fuse saving	LOST: F633 melts (0.100 s) before R2 fast (0.121 s)	HELD

Case 07 - LG fault at 646 (phase c), $Z_f = 0$ ohm (bolted), DG 4.05 MVA connected at 692 (Fig. 14 / 17 cell)

Single setting: R2 non-directional, forward setting only

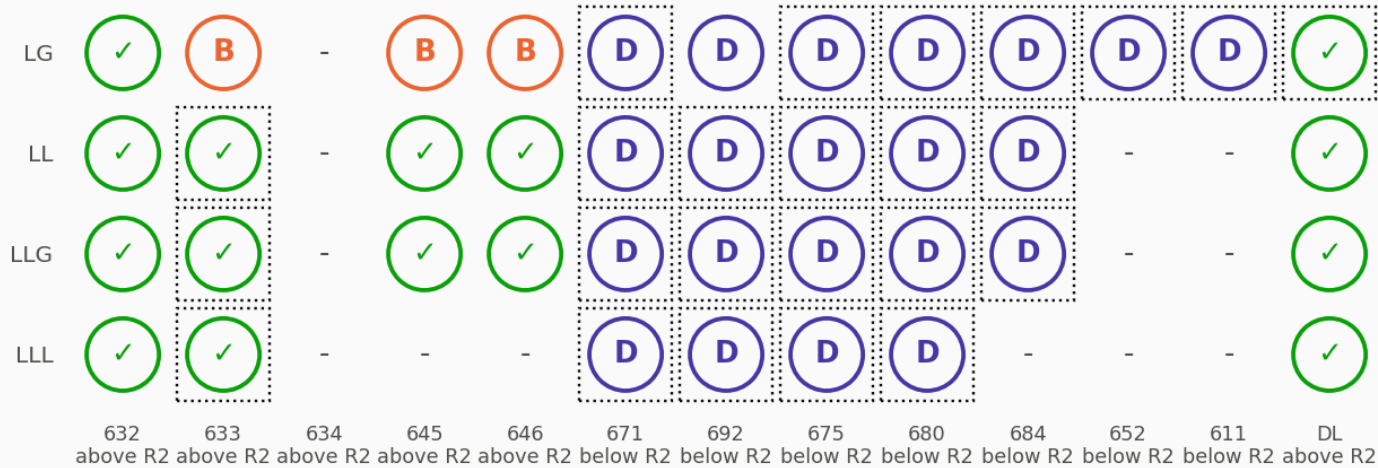


Dual setting: R2 as DSDR

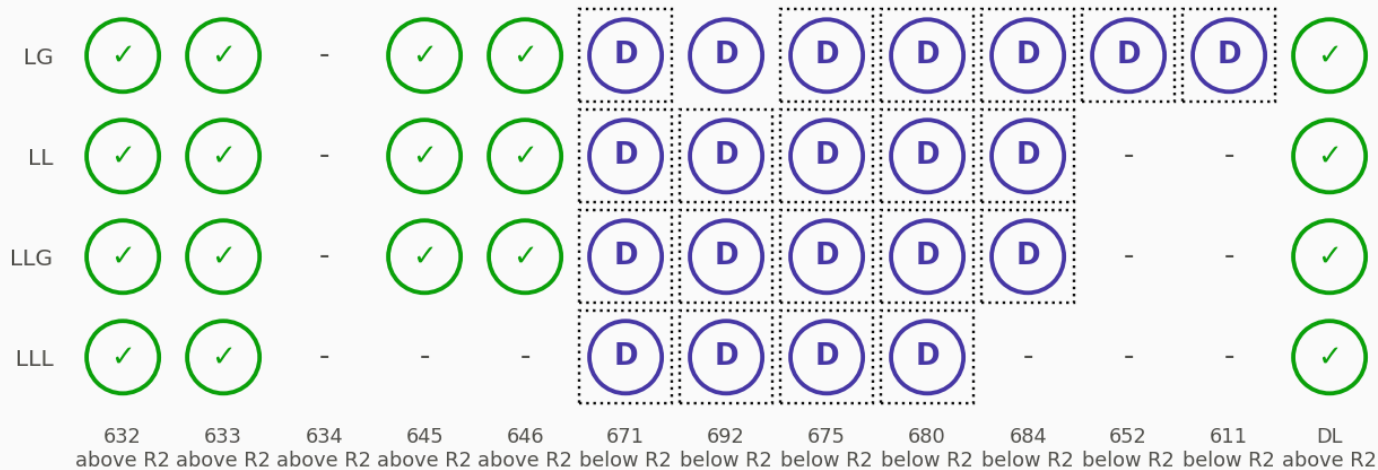


Quantity	Single setting	Dual setting (DSDR)
LG fault at 646 (phase c)	LG, phases C	(same fault and currents)
Fault impedance Z_f	0 ohm (bolted)	(same fault and currents)
Fault-path current (F646)	2732 A	(same fault and currents)
Current through R1, direction	2359 A, forward (RG60 -> 632), grid contribution	(same fault and currents)
Current through R2, direction	469 A, REVERSE (671 -> 632), DG contribution only	(same fault and currents)
R2 setting applied (fast / delayed)	FORWARD: plug 300/600 A, starts 600/1200 A, TMS 0.1/1.0	REVERSE: plug 150/300 A, starts 300/600 A, TMS 0.1/1.0
R1 fast / delayed	0.255 s / 5.105 s	0.255 s / 5.105 s
R2 fast / delayed	no trip / no trip	0.396 s / no trip
Primary fuse melts / clears	F646 400E: 1.057 s / 2.058 s	F646 400E: 1.057 s / 2.058 s
Backup fuse melts	F632 500E: 2.913 s	F632 500E: 2.913 s
Fuse saving	LOST: R2 does not pick up (469 A)	HELD

Single setting on R2 - time sequence: above R2 15 of 18 held; below R2 the DG keeps feeding 21 of 21



Dual setting on R2 (DSDR) - time sequence: above R2 18 of 18 held; below R2 the DG keeps feeding 21 of 21



✓ held: fault cut off, fuse below 100 % heat
 O fast sequence fine, a delayed-trip / series condition fails
 D below R2: the DG keeps feeding the fault

B R2 does not trip on the DG's reverse current
 X fuse melts before the reclosers finish
 ... differs from the strict rule

Fuse-saving coordination with the DG connected

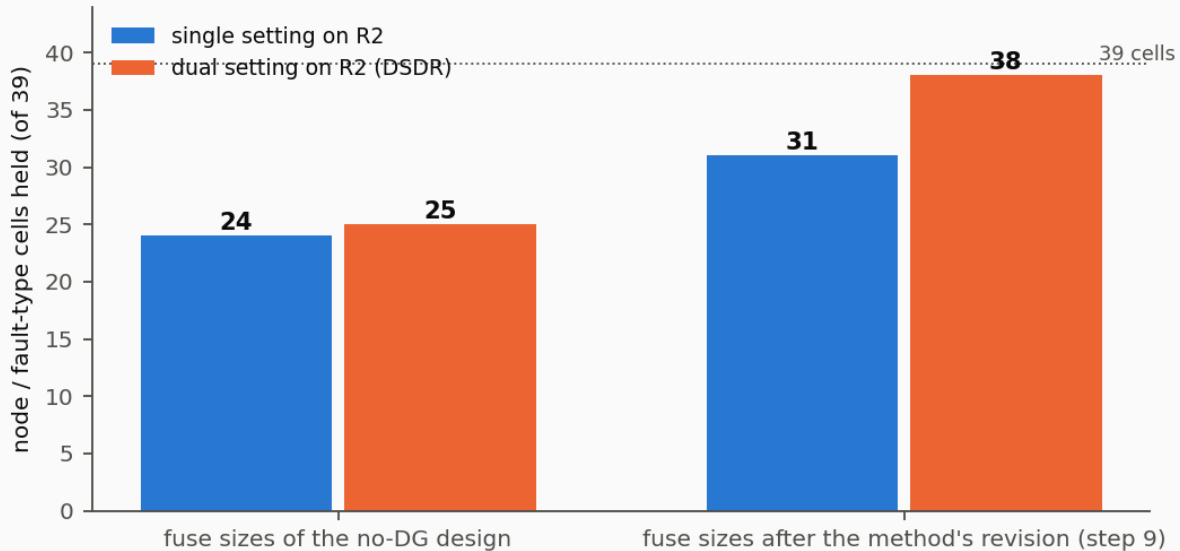


Fig. 1 - conventional recloser-fuse coordination for the fuse-saving scheme

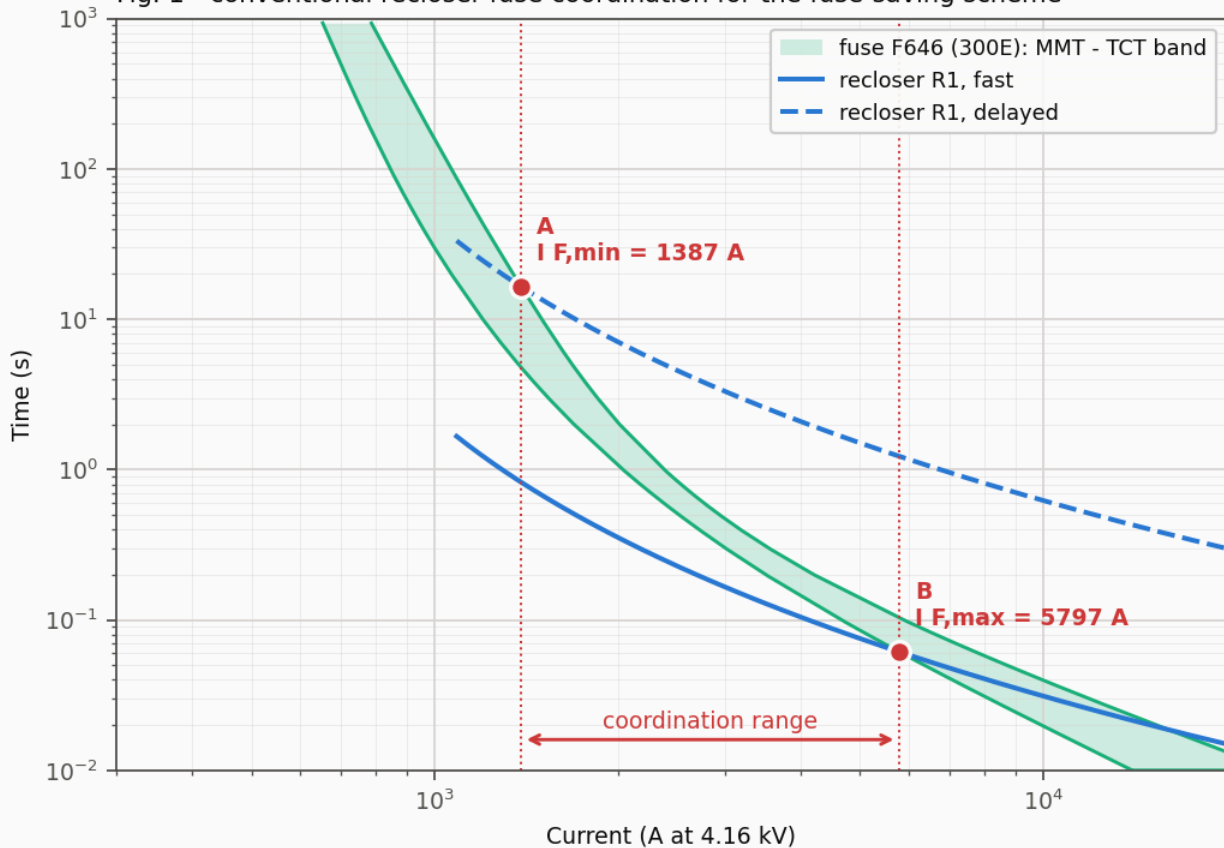


Fig. 3 - time-current characteristics of the dual-setting recloser R2: reverse and forward setting

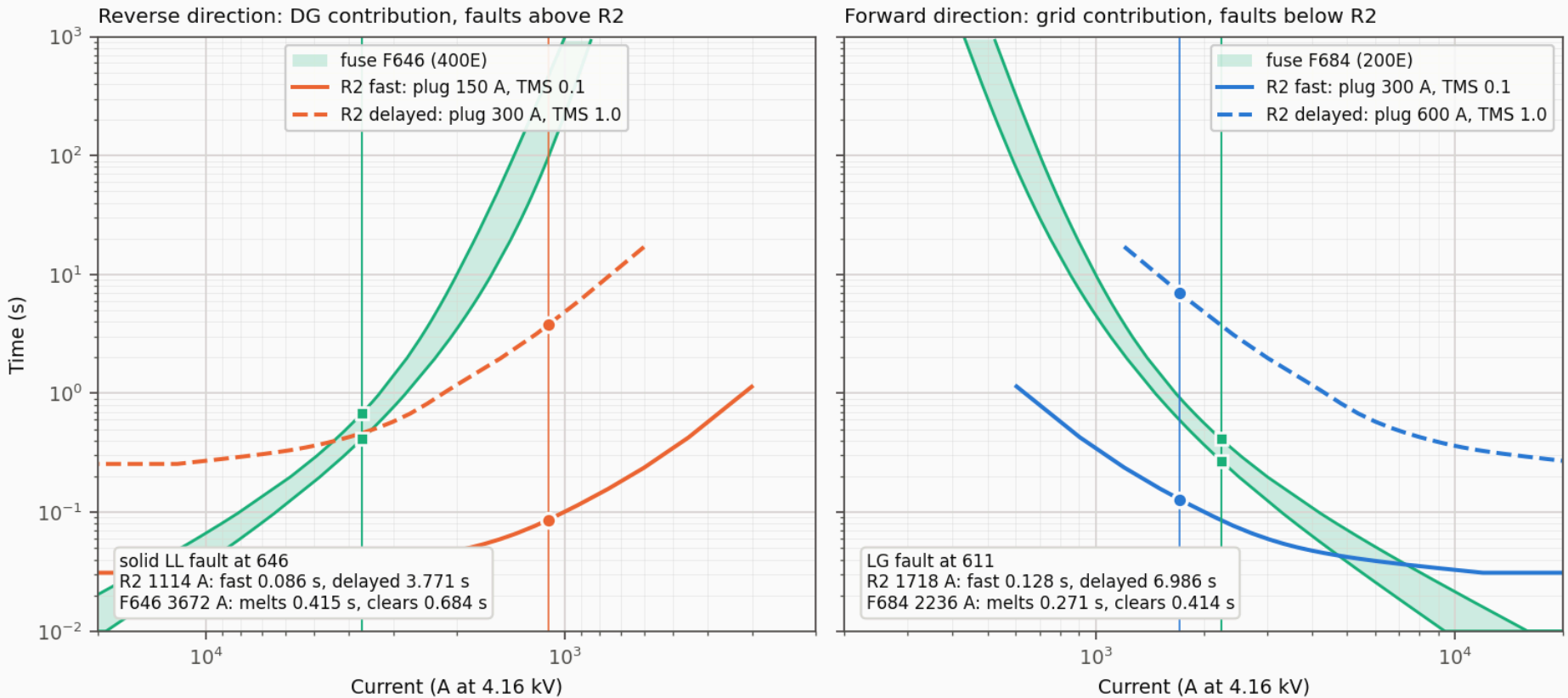


Fig. 8 LG fault at 611 - coordination held

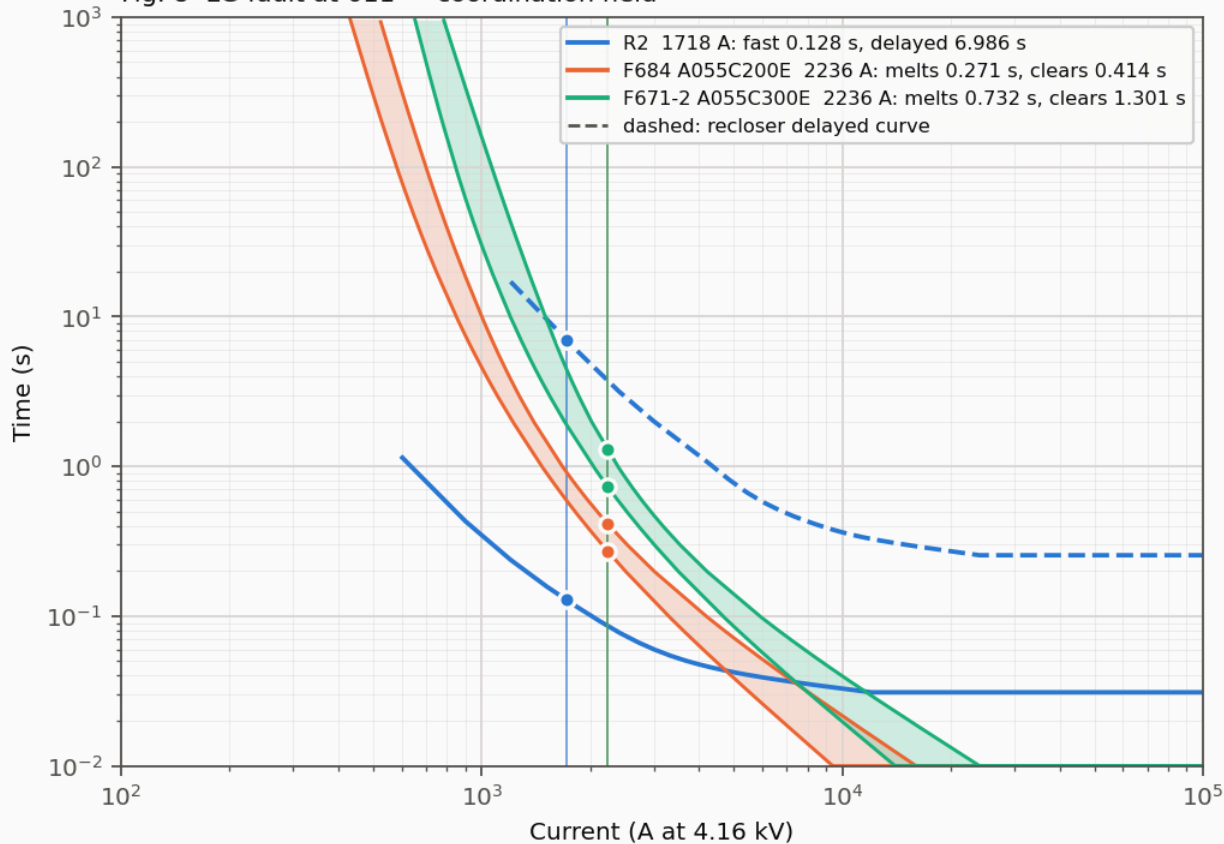


Fig. 9 LLG fault mid 692-675, 1 ohm

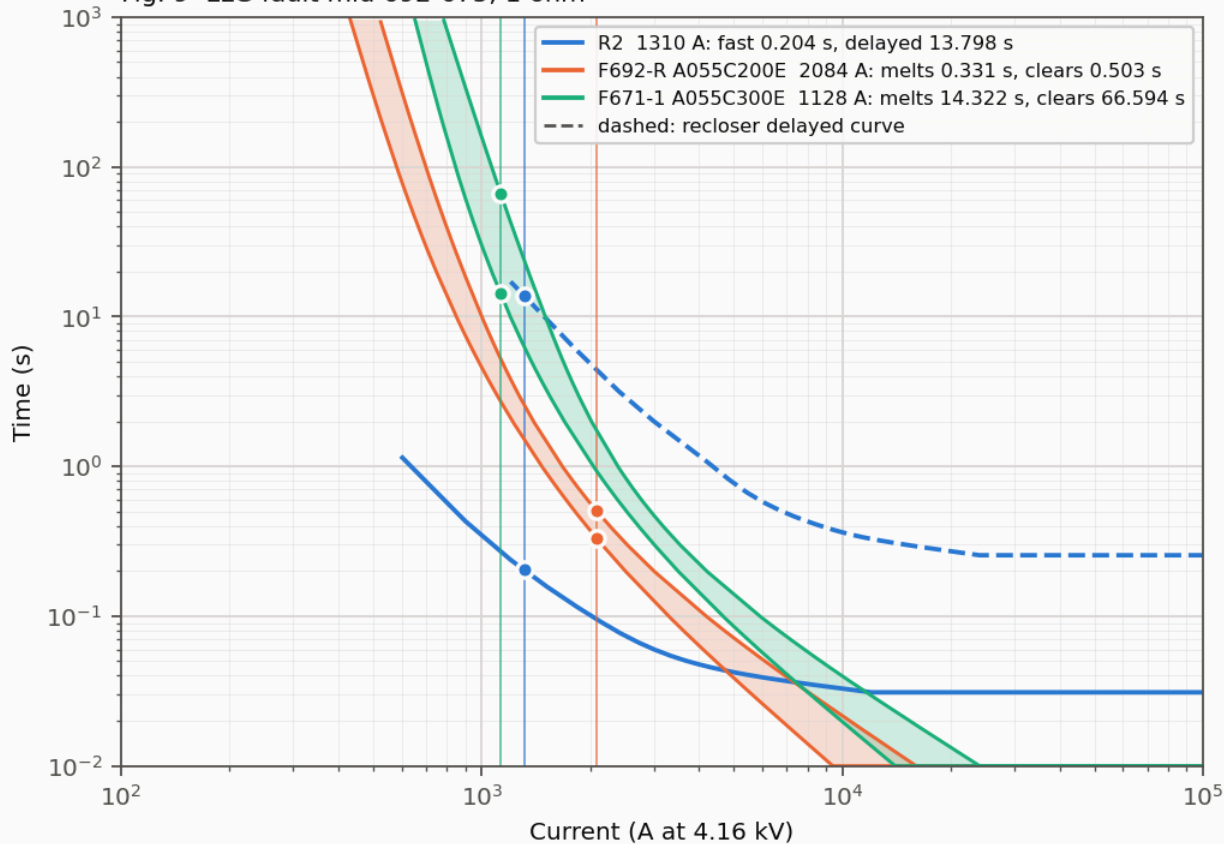


Fig. 10 - current at node 632, LL a-c fault at 684 through 0.2 ohm, PowerFactory EMT

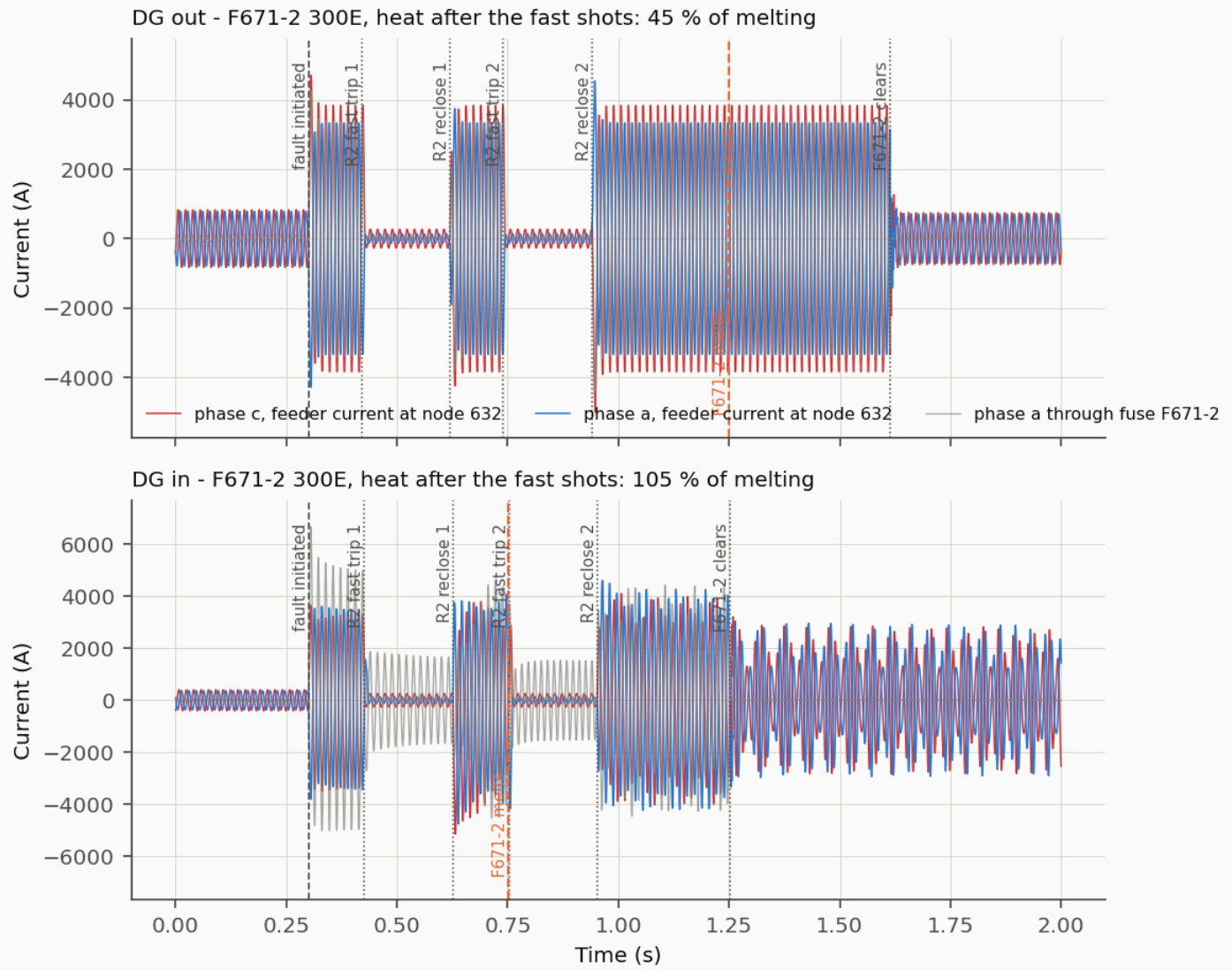


Fig. 11 LL fault at 646, 1 ohm - coordination held

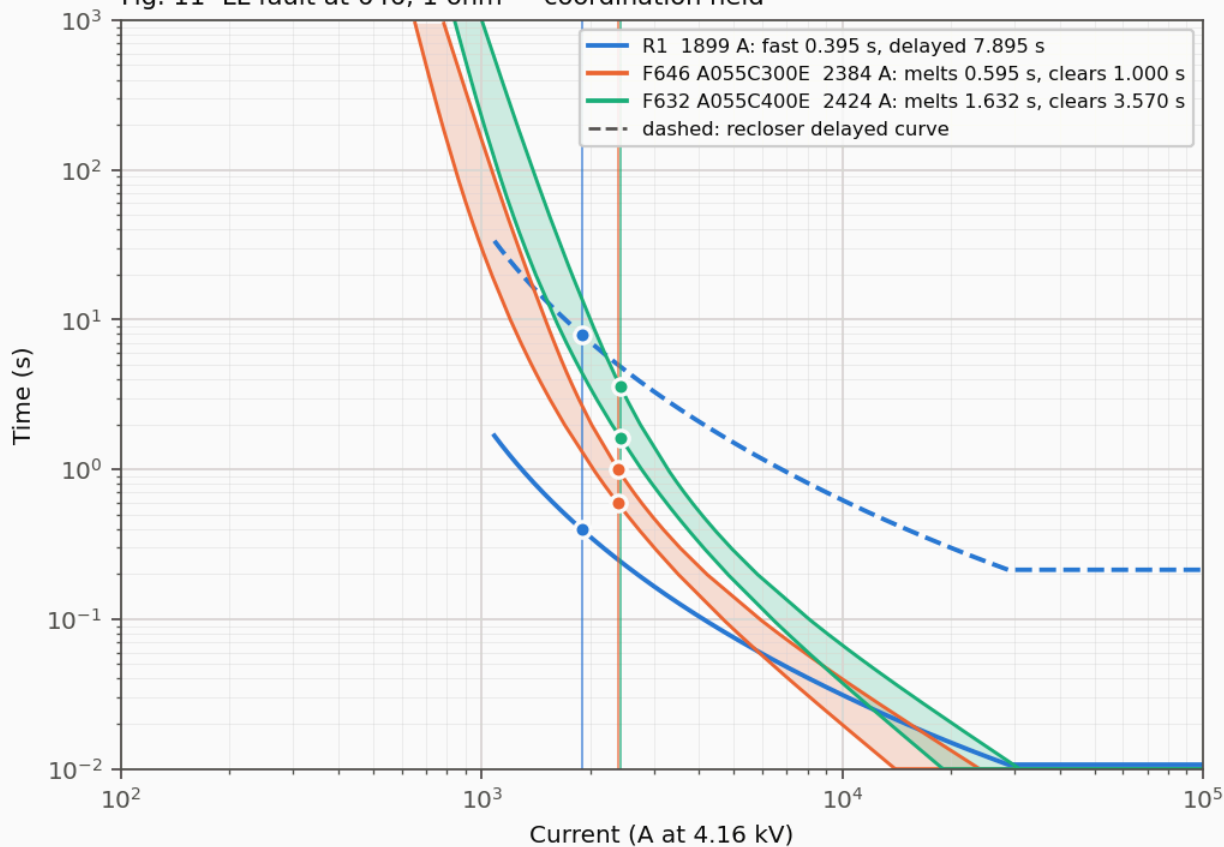


Fig. 12 LL at 645, 1.5 ohm, conventional R2 - coordination held

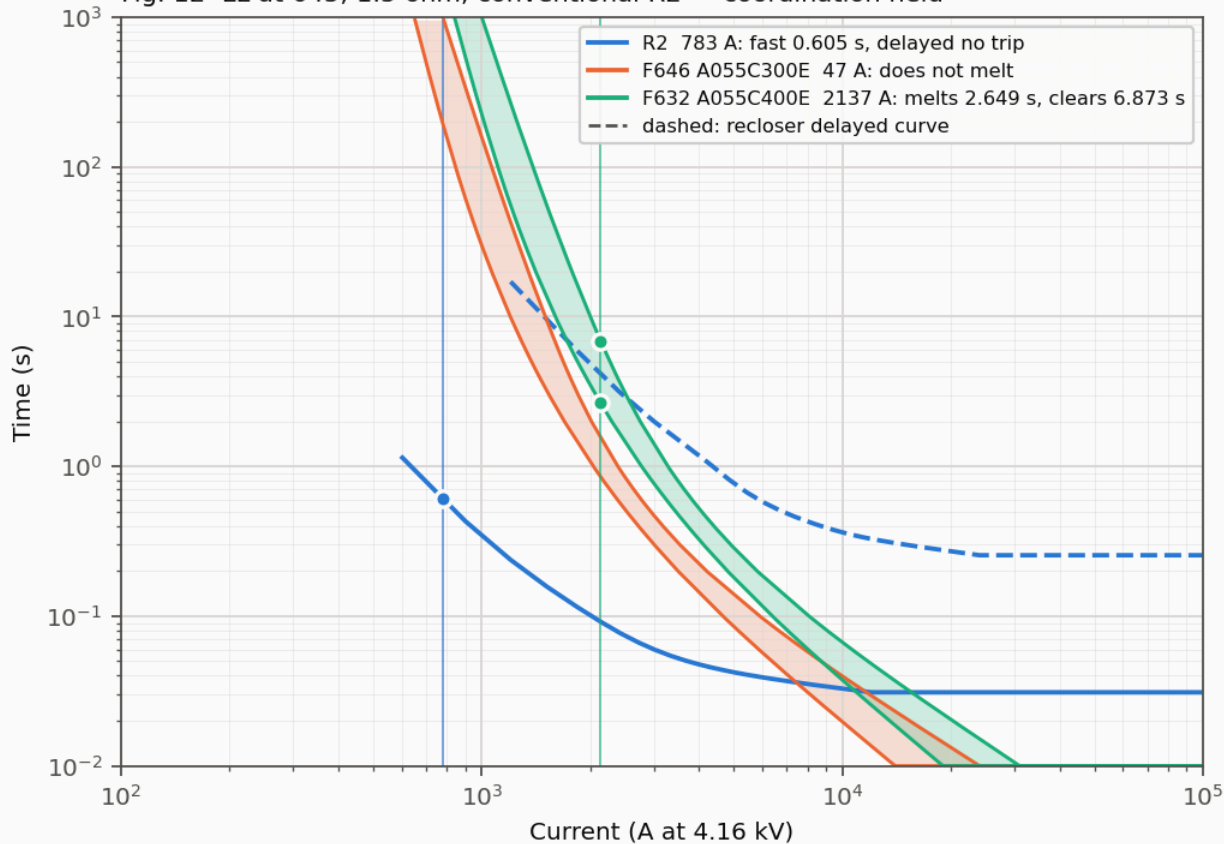


Fig. 13 3-ph fault at 10 % of 632-633, conventional R2

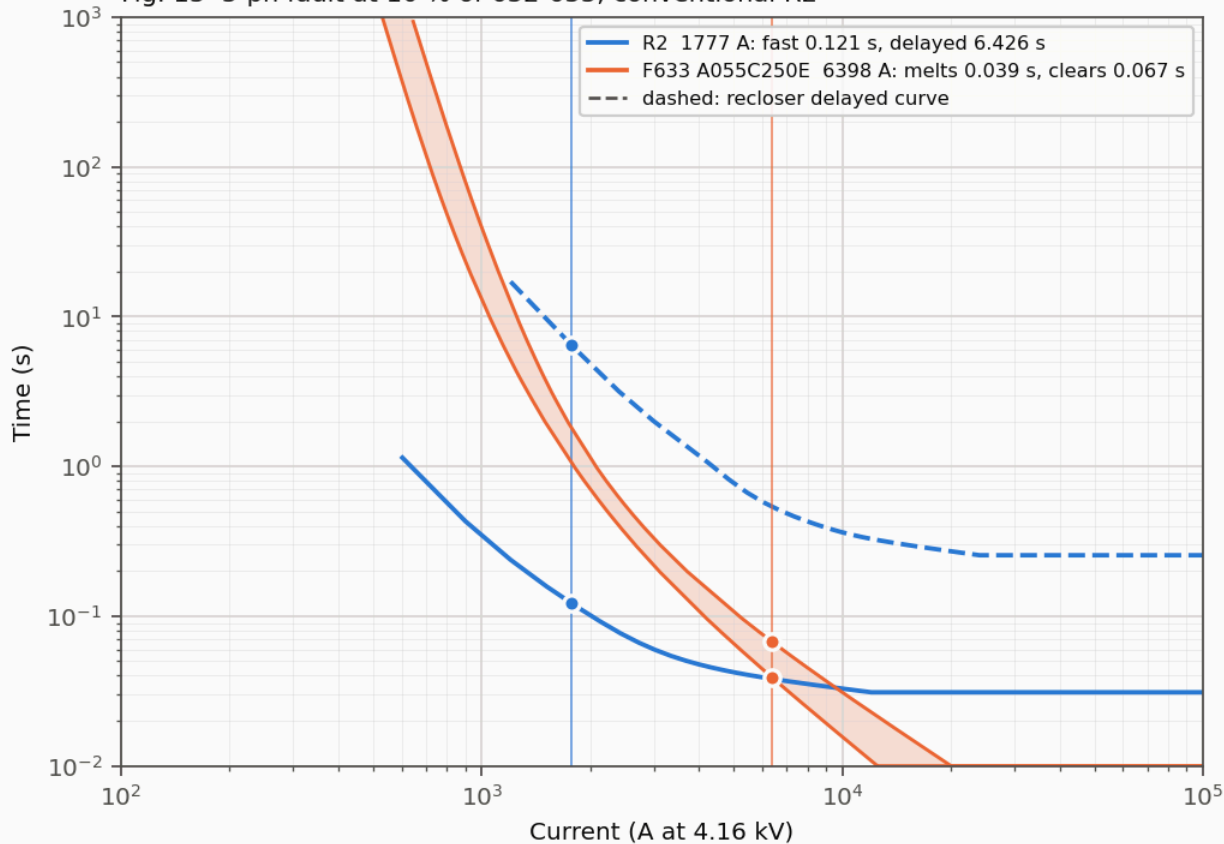


Fig. 14 - coordination without DSDR, DG 4.05 MVA at 692

LG			—										
LL			—								—	—	
LLG			—								—	—	
LLL			—	—	—					—	—	—	
	632	633	634	645	646	671	692	675	680	684	652	611	DL

Faulted node



coordination held



coordination lost



not applicable

Fig. 15 solid LL at 646, R2 as DSDR - coordination held

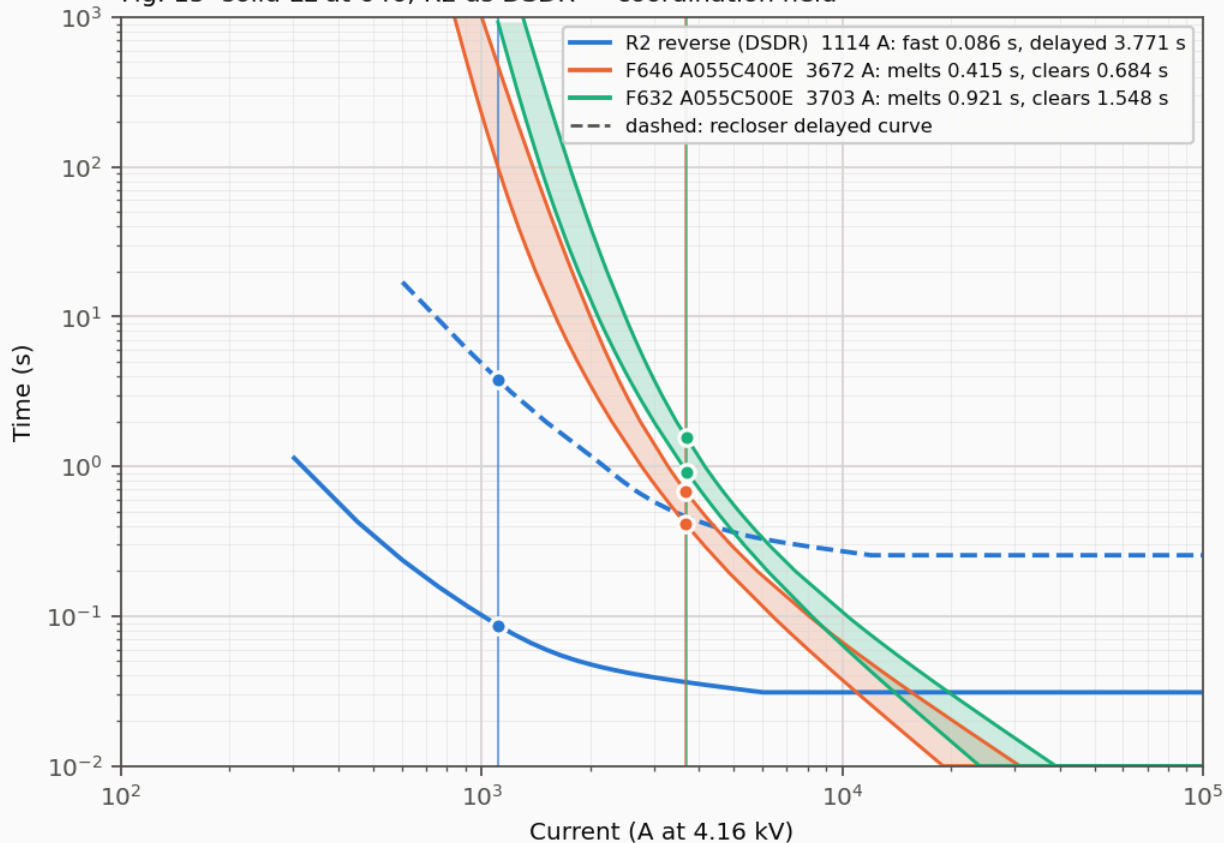


Fig. 16 solid LL at 646, R1 - coordination held

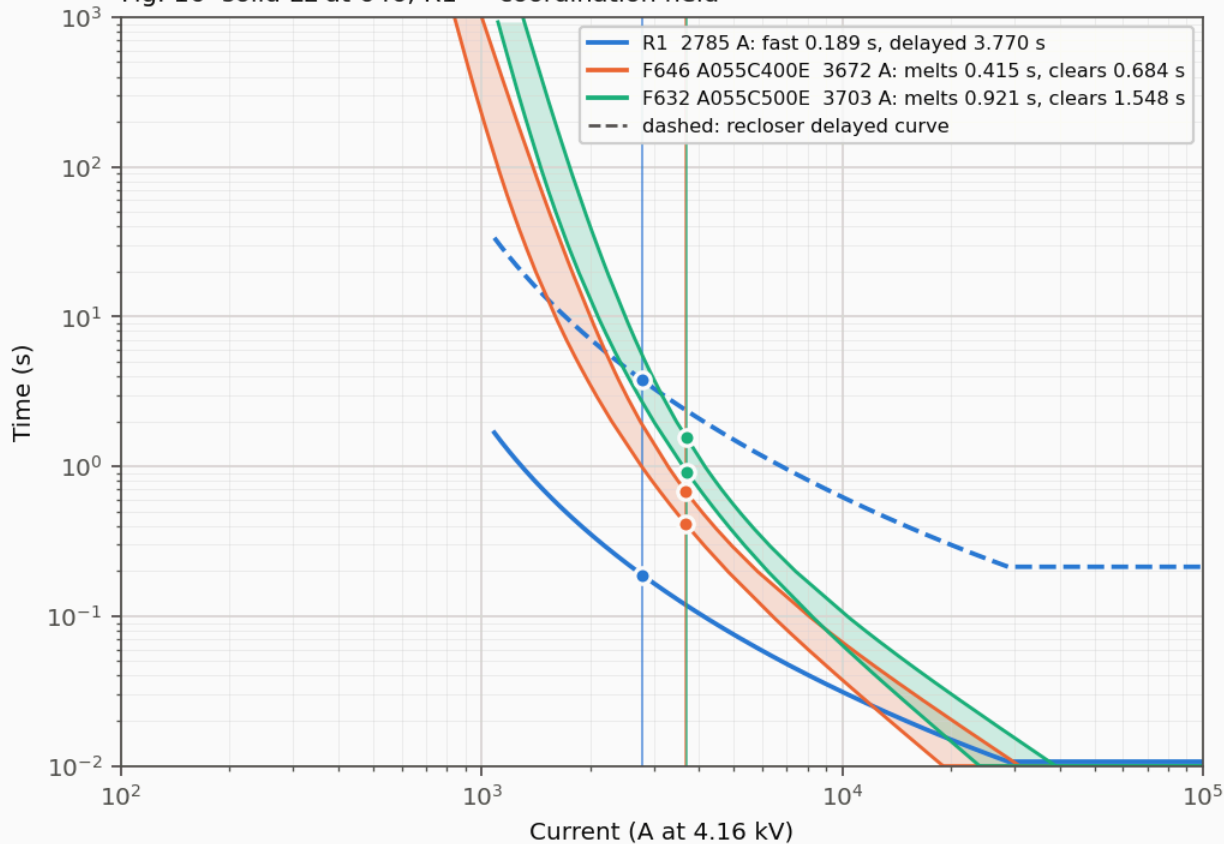


Fig. 17 - coordination with DSDR, DG 4.05 MVA at 692

LG			—										
LL			—								—	—	
LLG			—								—	—	
LLL			—	—	—					—	—	—	
	632	633	634	645	646	671	692	675	680	684	652	611	DL

Faulted node



coordination held

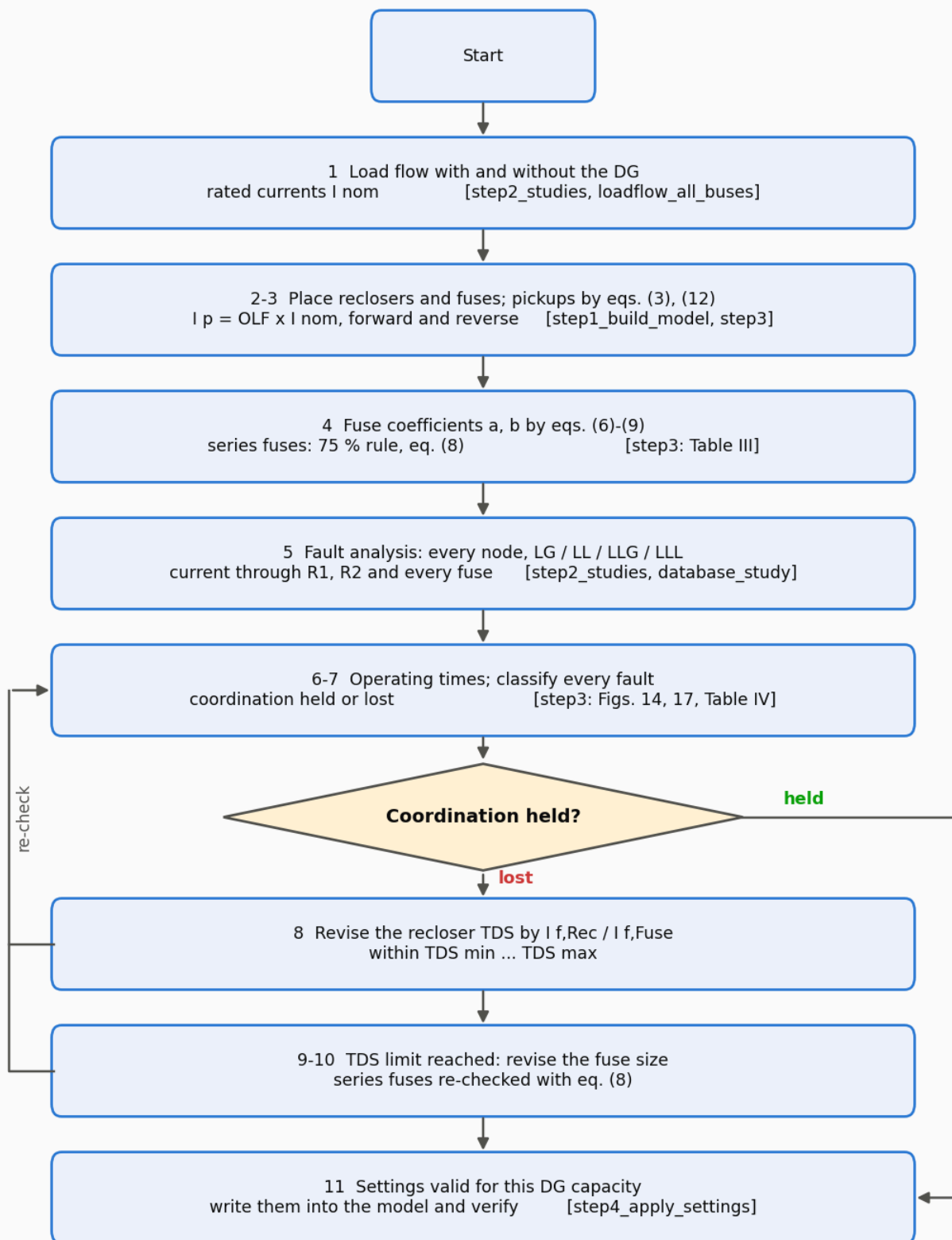


coordination lost

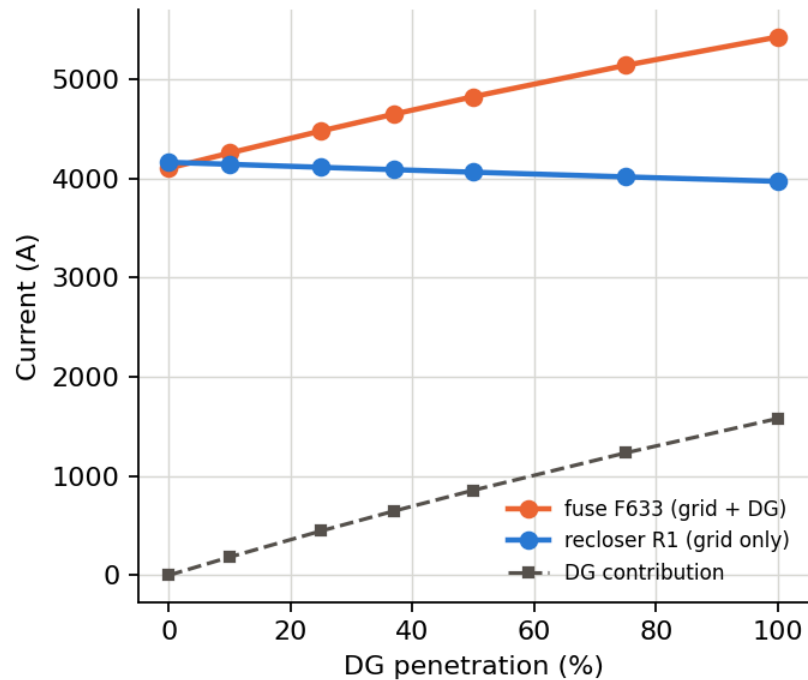


not applicable

Fig. 5 - the method for recloser-fuse coordination with the DSDR, with the script of each step



Fault LLL at 633: currents



Fault LL at 684 (lateral from 671): currents

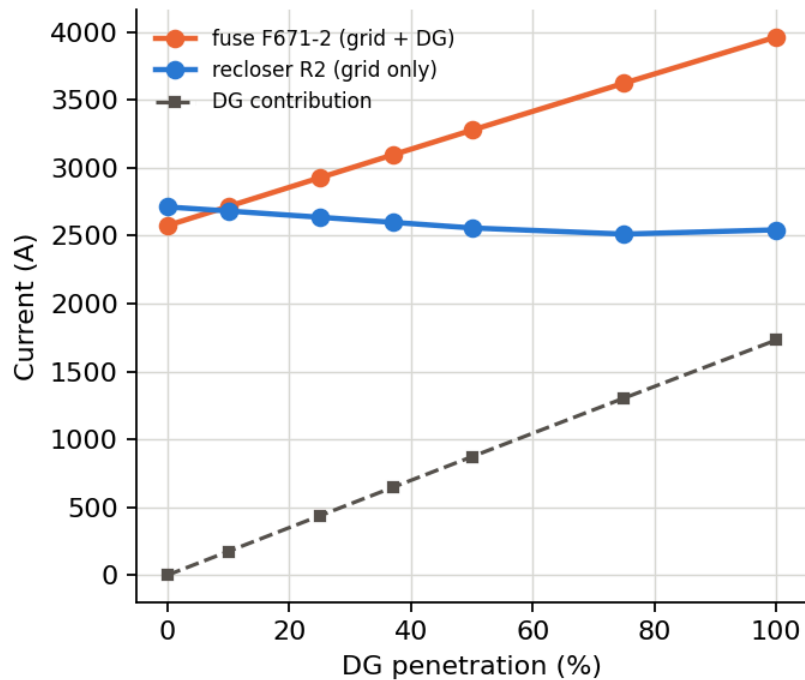
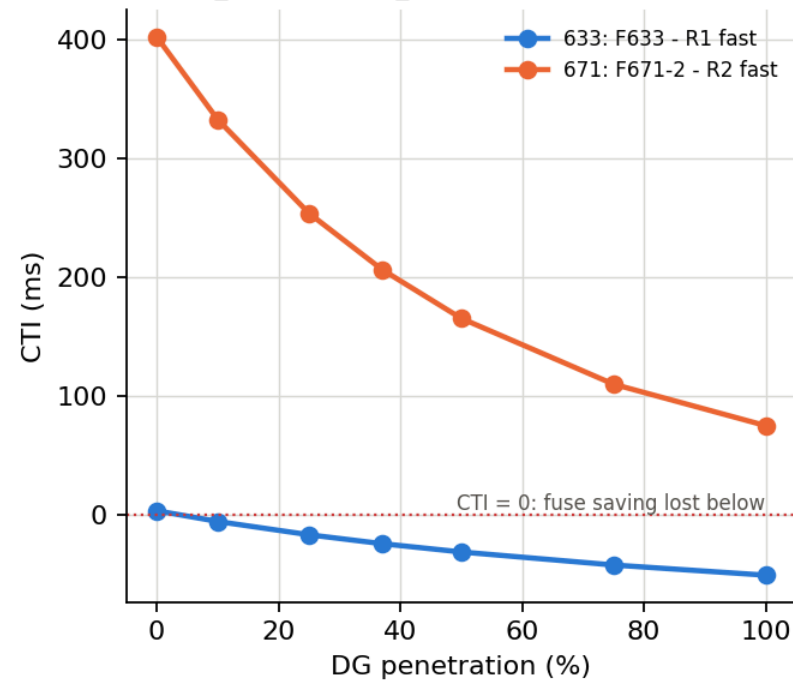
CTI = $t_{\text{MMT}}(\text{fuse}) - t_{\text{fast}}(\text{recloser})$ 

Fig. 7 - percentage variation of the CTI with DG penetration (each ring = 100 %; negative = fuse melts b

PowerFactory, conventional protection

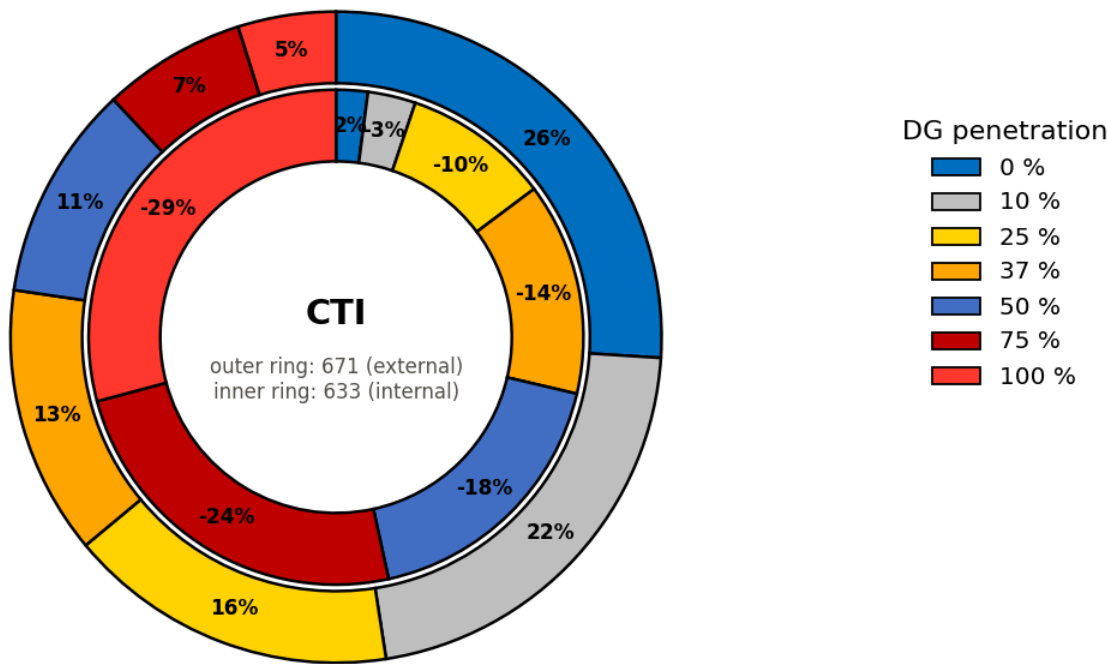
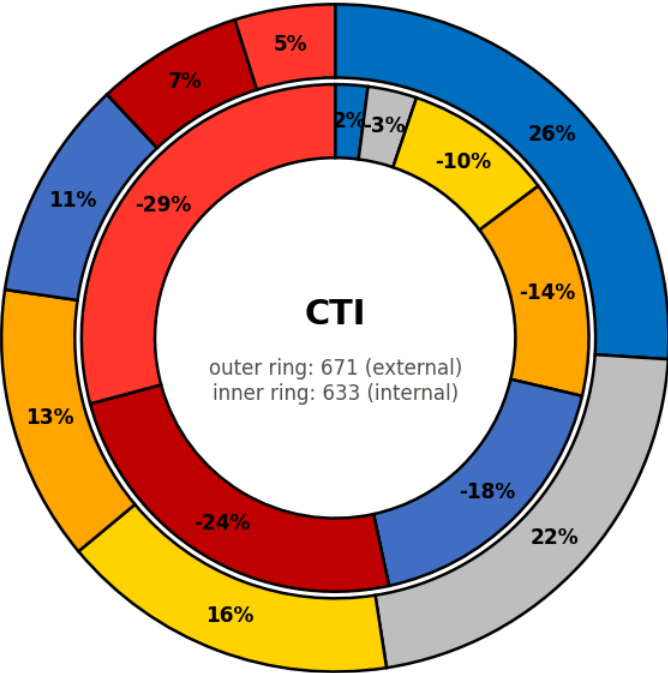
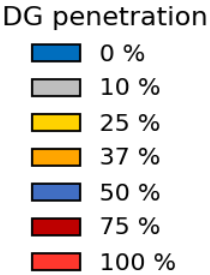
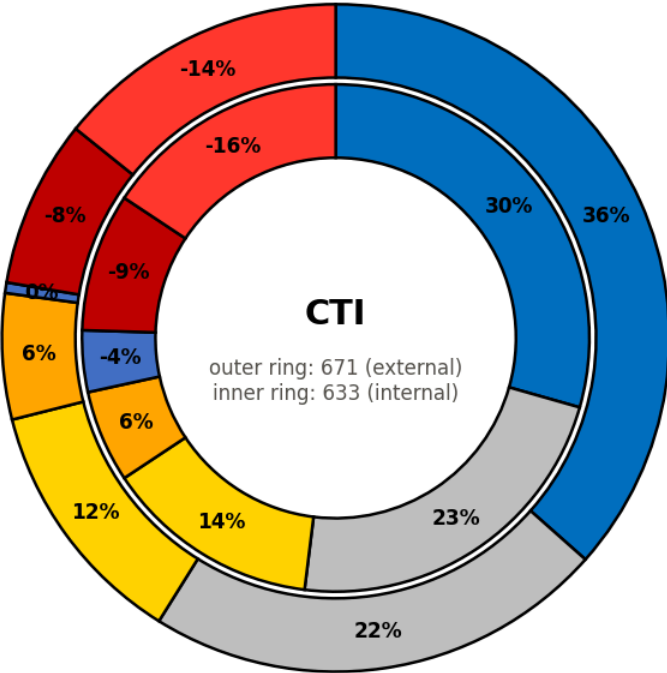


Fig. 7 - percentage variation of the CTI with DG penetration (each ring = 100 %; negative = fuse melts before the recloser's fast trip)

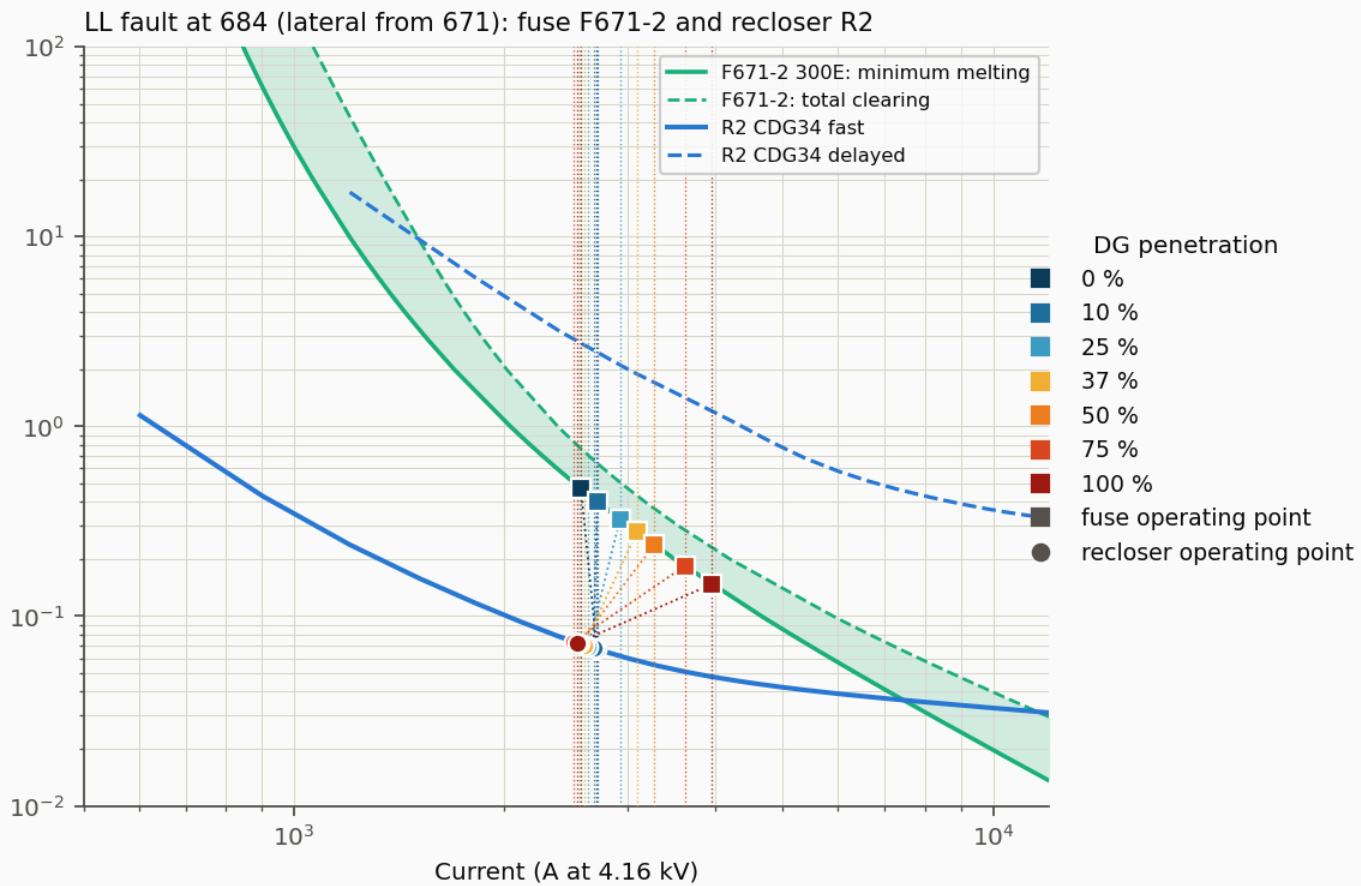
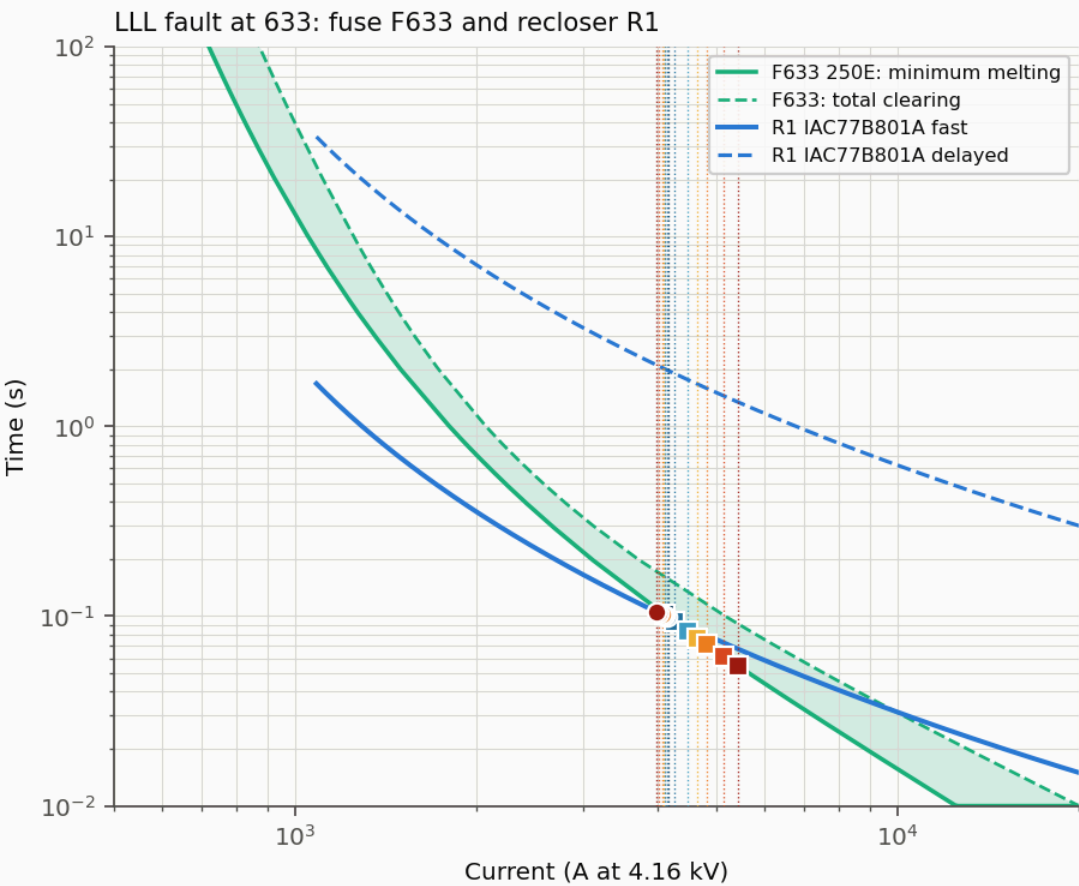
This study (PowerFactory)



Reference paper (values read from its Fig. 7)



TCC with increasing DG penetration (conventional settings): the recloser point stays, the fuse point moves right and down



TCC - LLL fault at 633: fuse F633 and recloser R1, conventional settings, at each DG penetration level (square: fuse, circle: recloser)

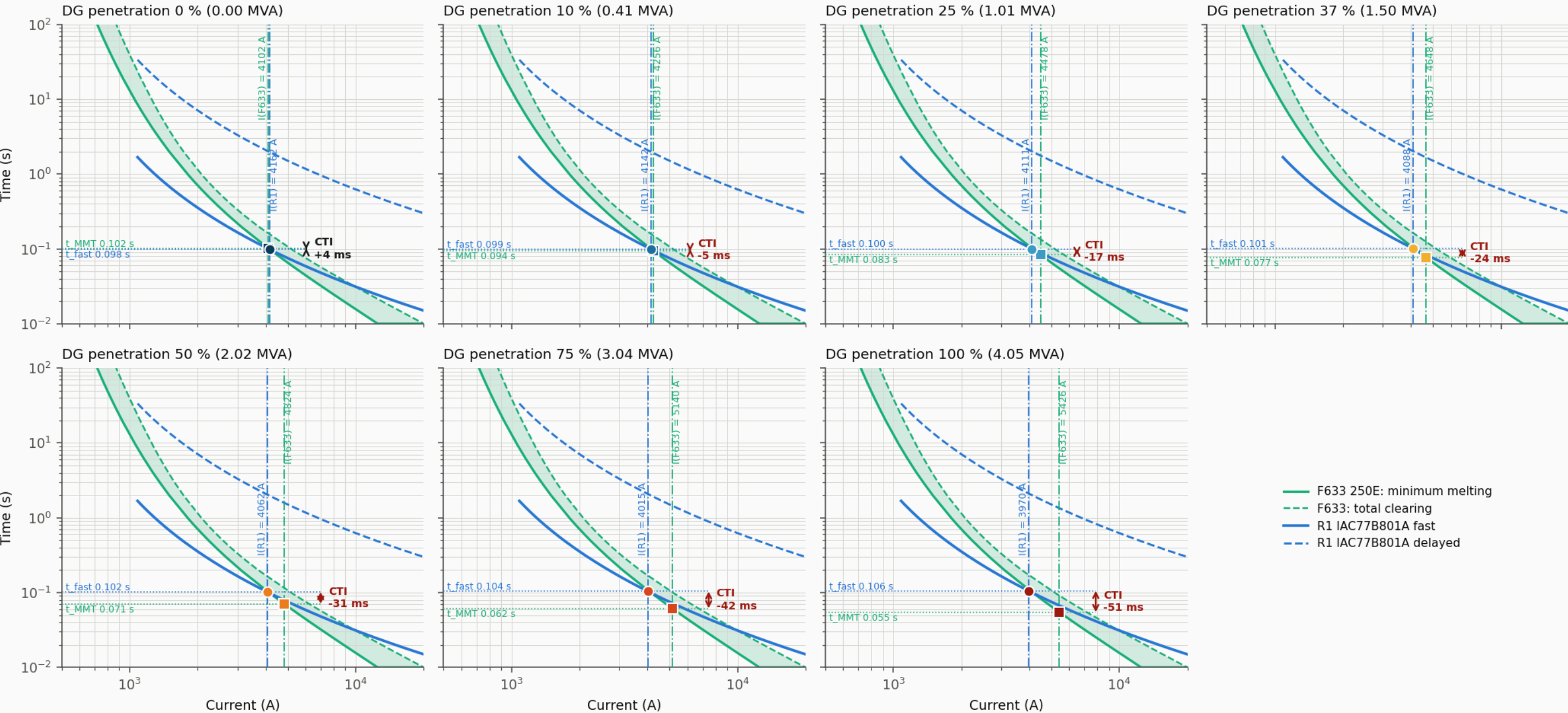


Fig. 6 - IEEE 13-node test feeder with the protective devices of this study

